

Quest4Knowledge

Team Captain Name: P.Krishna Koushik

Team Captain Mobile Number: 8019435543

Team Captain Email: 23h51a66h2@cmrcet.ac.in

No. of Team Members:3

Mentor Name (if any) :

Team Members' Details (if any):

i)Kavyasri Chowdary

ii)Snehal

College Name:CMRCET



Domain and Problem Statement

- Category: EduTech
- Problem Statement 4: Gamified Learning for Rural Education

Description: Students in rural areas often face challenges such as a lack of access to quality educational resources, limited exposure to engaging learning methods, and reduced motivation to pursue studies. Design a gamified learning platform for core subjects that uses elements like quizzes, rewards, progress tracking, and storytelling to make education interactive, fun, and accessible. The system should be lightweight, multilingual, and compatible with low-end devices to ensure inclusivity.

Requirements: Game development frameworks (Unity, Godot, or HTML5/JavaScript-based engines) for creating interactive modules, along with mobile/web technologies (Android SDK, Flutter, React, and Node.js) to ensure accessibility, offline support, and multilingual features.

Expected Outcomes:

- A gamified learning application (mobile/web) tailored for rural students with engaging modules for core subjects (like math, science, or language learning) with levels, badges, and leaderboards.
- Offline functionality or low-data usage features to ensure accessibility



Idea/Solution



We propose a **hybrid gamified-learning FPS platform** that fuses immersive action gameplay with rigorous academic content. Players navigate a stylized environment (e.g. futuristic arena, forest, village-themed backdrop), discover or trigger floating question panels or “targets,” and then **shoot the correct answer** among multiple choice options. Correct hits reward points, XP, badges, and unlock new levels or mini-story arcs. Wrong shots prompt hints, retries, or short remediation popups.

But it's not just a shooter:

- Quizzes, fill-in-the-blanks, drag-and-drop puzzles, mini logic games, and interactive simulations are interlaced between FPS segments.
- After rounds, players can review explanations, get tips, retry missed questions, and see progress dashboards.
- Storytelling ties modules together: e.g. a student character ventures through villages, unlocking “knowledge gates” in math, science, language.
- Multilingual support, optional voiceover narration, accessible UI, and offline caching ensure it works even on low-end devices in rural areas.
- This is more engaging than a plain quiz app, and more pedagogically sound than a pure shooter: it **bridges gaming and learning** in a coherent, modular structure.



Existing Vs Proposed System

- Most current gamified learning apps rely on simple points, badges, and leaderboards. While studies show these features can improve engagement, they often remain limited to 2D quizzes, flashcards, or videos. Many require constant internet connectivity, high-end devices, and use heavy data, making them unsuitable for rural or low-resource settings. Multilingual options exist, but localization is usually shallow and not tailored to cultural or regional contexts. Content is also rigid, with little scope for teachers to add new material. Feedback is mostly given after the quiz, with limited opportunities for real-time guidance.
- Our solution goes further by combining real-time FPS-style interaction with quizzes for deeper engagement. Students can actively shoot correct answers in an immersive 3D environment, enhanced by dynamic visuals and immediate feedback. The system supports offline caching, compressed assets, and adaptive graphics to run on both high-end and low-end devices. We also focus on deep localization with local dialects, voiceovers, and culturally relevant examples. The modular architecture allows educators to add or adapt content easily. Finally, students receive real-time hints, retries, and adaptive difficulty to ensure meaningful learning and not just scoring.



Motivation

Persistent Rural Education Gaps

Students in rural areas often lack access to quality resources, suffer from teacher shortages, and face infrastructure constraints.

Motivation & Engagement Deficit

Traditional rote methods often lead to disengagement, high dropout rates, and lower retention, especially among marginalized populations.

Digital Divide Constraints

Even when edtech is available, low bandwidth, intermittent connectivity, and low-end hardware limit adoption.

Scalability & Equity

A solution that works across multiple geographies, languages, and resource constraints can democratize access to quality education.

Evidence of Gamification Benefits

Research shows gamified systems can change student behavior, improve accuracy of responses, and bolster motivation when implemented well



Objective & Scope of Solution

Main Goals

- Develop a **lightweight, modular learning game** that blends FPS mechanics with quizzes and simulations.
- Ensure **offline access, multilingual support, and adaptive difficulty** for inclusivity.
- Provide **teacher and admin tools** for content creation, analytics, and progress tracking.
- Guarantee **cross-platform compatibility** (Android, web, desktop) with smooth fallback for low-end devices.
- Deliver **engaging gameplay** that motivates students to learn voluntarily.

Boundaries

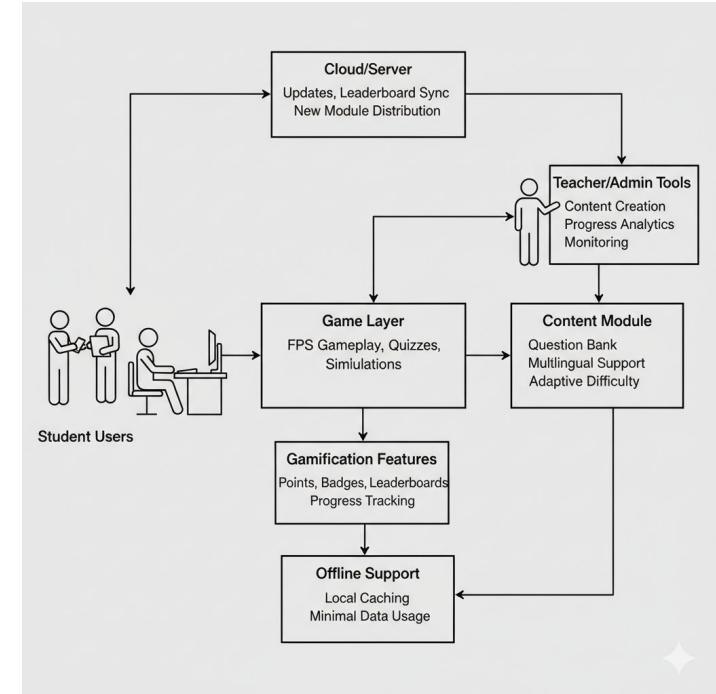
- Prototype/demo limited to short **8-second clips** for hackathon presentation.
- Focus on **core mechanics** over full commercial polish.
- Graphics must **scale across devices** with low/medium/high settings.



System Architecture

High-Level Design & Working Flow

- **Student Users:** Access the platform via mobile, tablet, or desktop.
- **Game Layer:** Core engine combining FPS gameplay, quizzes, and simulations for interactive learning.
- **Gamification Features:** Adds motivation with points, badges, progress tracking, and leaderboards.
- **Content Module:** Stores the question bank, supports multilingual delivery, and adapts difficulty based on student performance.
- **Teacher/Admin Tools:** Enable educators to create content, track progress, and monitor analytics.
- **Offline Support:** Local caching ensures smooth learning with minimal data usage, even in low-connectivity areas.
- **Cloud/Server:** Manages updates, synchronizes leaderboards, and distributes new content modules





Resources Required

Game / Engine

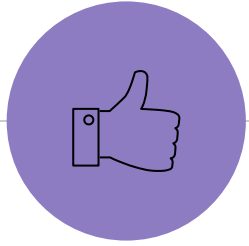
- Unreal Engine 5 (Lumen, Nanite, scalable rendering; fallback for low-end devices).
- Alternative lightweight 3D engine if required.

Content / Database

- Web admin panel for teachers (quiz upload, content mgmt).
- Analytics (progress, error tracking).

Backend / Infra

- Node.js + Express or serverless API for sync & analytics.
- Local caching (SQLite) for offline use.
- Cloud/CDN for updates & assets.



Thank You!